



Linear Programming: Model Formulation and Graphic Solution Chapter 2



Linear Programming

A Linear Programming model seeks to maximize or minimize a linear function, subject to a set of linear constraints.

The linear model consists of the following components:

A set of decision variables. An objective function.

A set of constraints.

Formulation of LPP

Structure of all LPP has three important components.

- (1) Decision variables (activities) : These are activities for which we want to determine a solution. These are usually denoted by $x_1, x_2, \dots, x_n$.
- (2) The objective function (goal) : This is a function which is expressed in terms of decision variables and we want to optimize (maximize or minimize) this function.
- (3) The constraints : These are limiting conditions on the use of resources. The solution of LPP must satisfy all these constraints.

Formulation of LPP

LP Model Formulation (cont.)

Max/min $z = c_1x_1 + c_2x_2 + \dots + c_nx_n$

subject to:

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n (\leq, =, \geq) b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n (\leq, =, \geq) b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n (\leq, =, \geq) b_m \end{array} \right.$$

x_j = decision variables

b_i = constraint levels

c_j = objective function coefficients

a_{ij} = constraint coefficients

A Maximization Model Example

Example : The Beaver Creek Pottery Company

Given these limited resources, the company desires to know how many bowls and mugs to produce each day in order to maximize profit.

There are 40 hours of labor and 120 kg of clay available each day for production.

Resource Requirements

Product	<i>Labor</i>	<i>Clay</i>	<i>Profit</i>
	(hr/unit)	(kg/unit)	(\$/unit)
Bowl	1	4	40
Mug	2	3	50

Modeling

Decision Variables

x_1 : number of bowls to produce

x_2 : number of mugs to produce

Objective Function

Total Profit = $40 x_1 + 50 x_2$

$40 x_1$ = profit from bowls

$50 x_2$ = profit from mugs

Maximize $Z = 40 x_1 + 50 x_2$

Model Constraints

Labor Constraint

Total Labor used in production = $1 x_1 + 2 x_2$

$$1 x_1 + 2 x_2 \leq 40 \text{ hr}$$

Clay Constraint

Total Clay used in production = $4 x_1 + 3 x_2$

$$4 x_1 + 3 x_2 \leq 120 \text{ lb}$$

Non-negativity Constraint

$$x_1 \geq 0, x_2 \geq 0$$

Linear Programming Model

Maximize $Z = 40 x_1 + 50 x_2$

Subject to:

$$1 x_1 + 2 x_2 \leq 40$$

$$4 x_1 + 3 x_2 \leq 120$$

$$x_1, x_2 \geq 0$$

Feasible / Infeasible

If $x_1 = 5$, $x_2 = 10$

$$Z = 40 \cdot 5 + 50 \cdot 10 = 700$$

If $x_1 = 2$, $x_2 = 2$

$$Z = 40 \cdot 2 + 50 \cdot 2 = 180$$

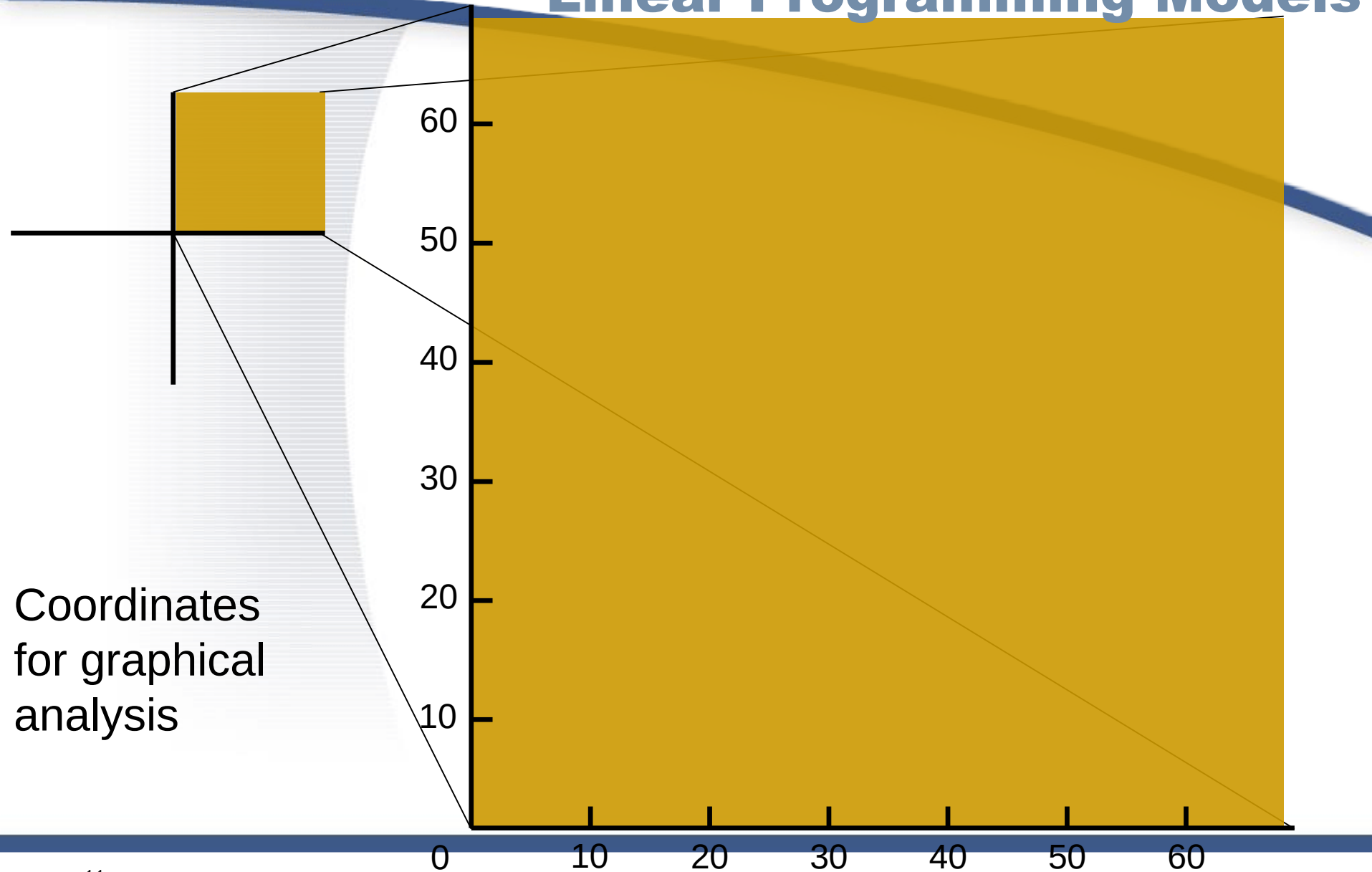
If $x_1 = 10$, $x_2 = 20$

$$Z = 40 \cdot 10 + 50 \cdot 20 = 1400$$

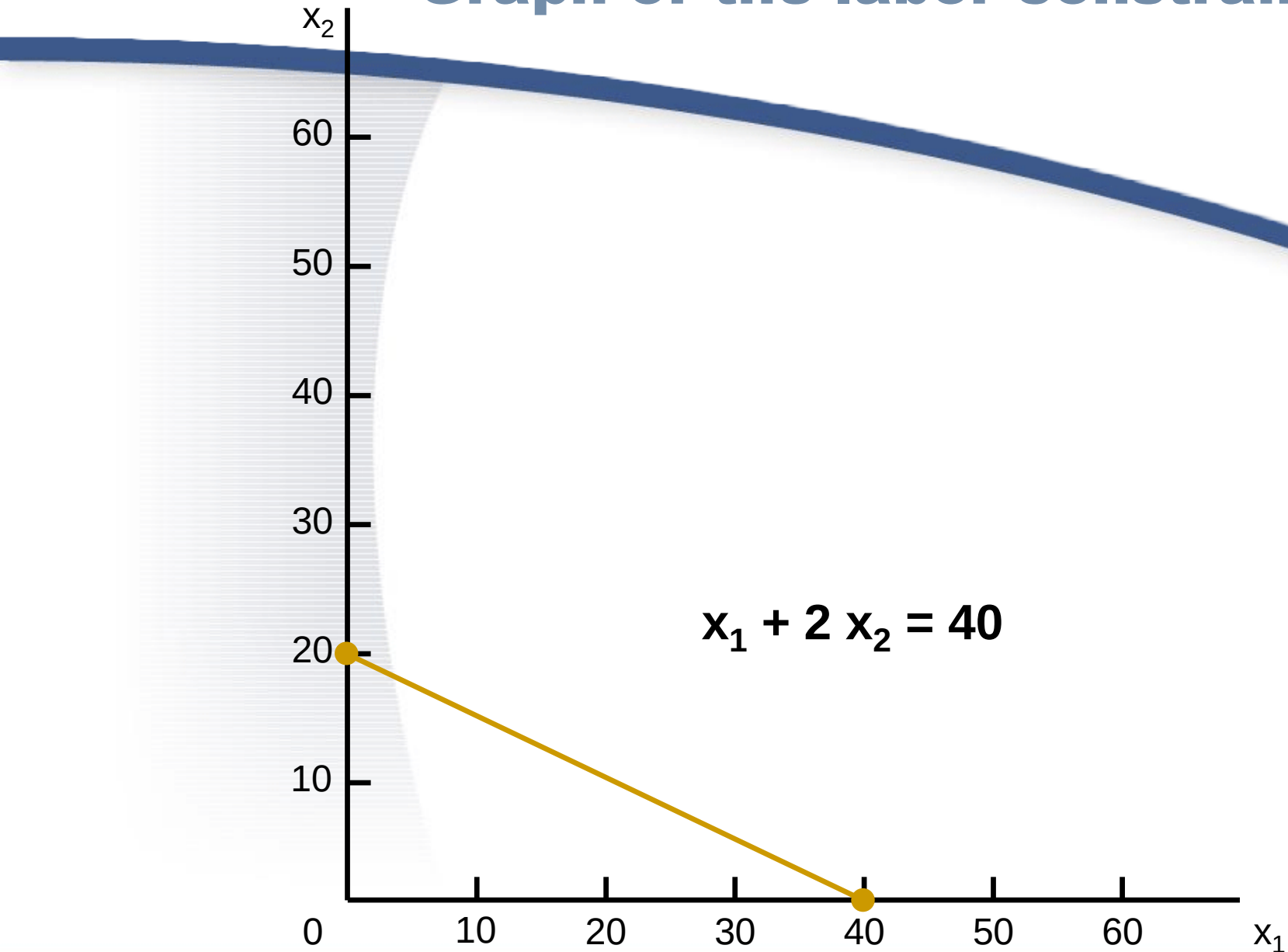
If $x_1 = 20$, $x_2 = 20$

$$Z = 40 \cdot 20 + 50 \cdot 20 = 1800$$

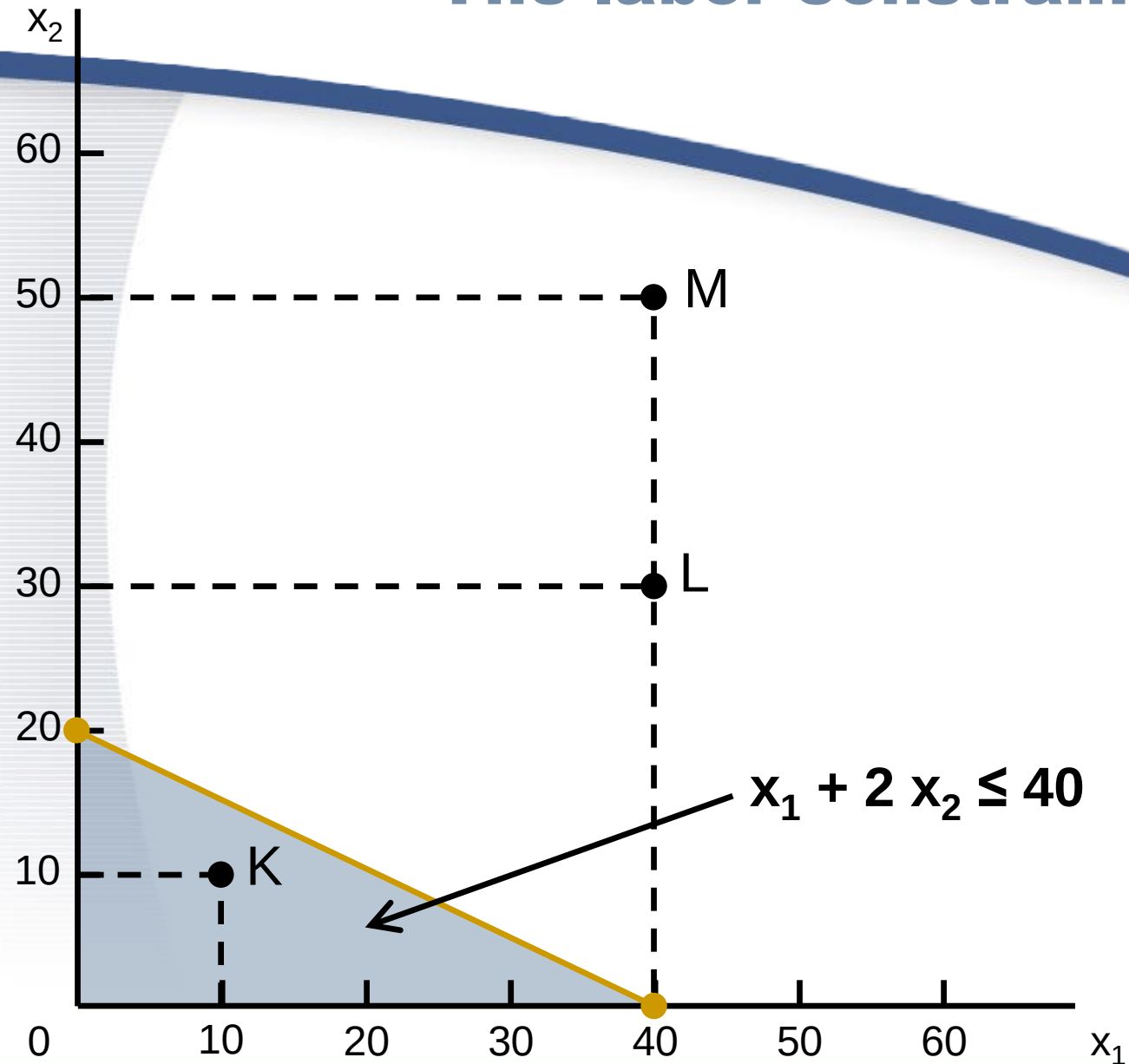
Graphical Solution of Linear Programming Models



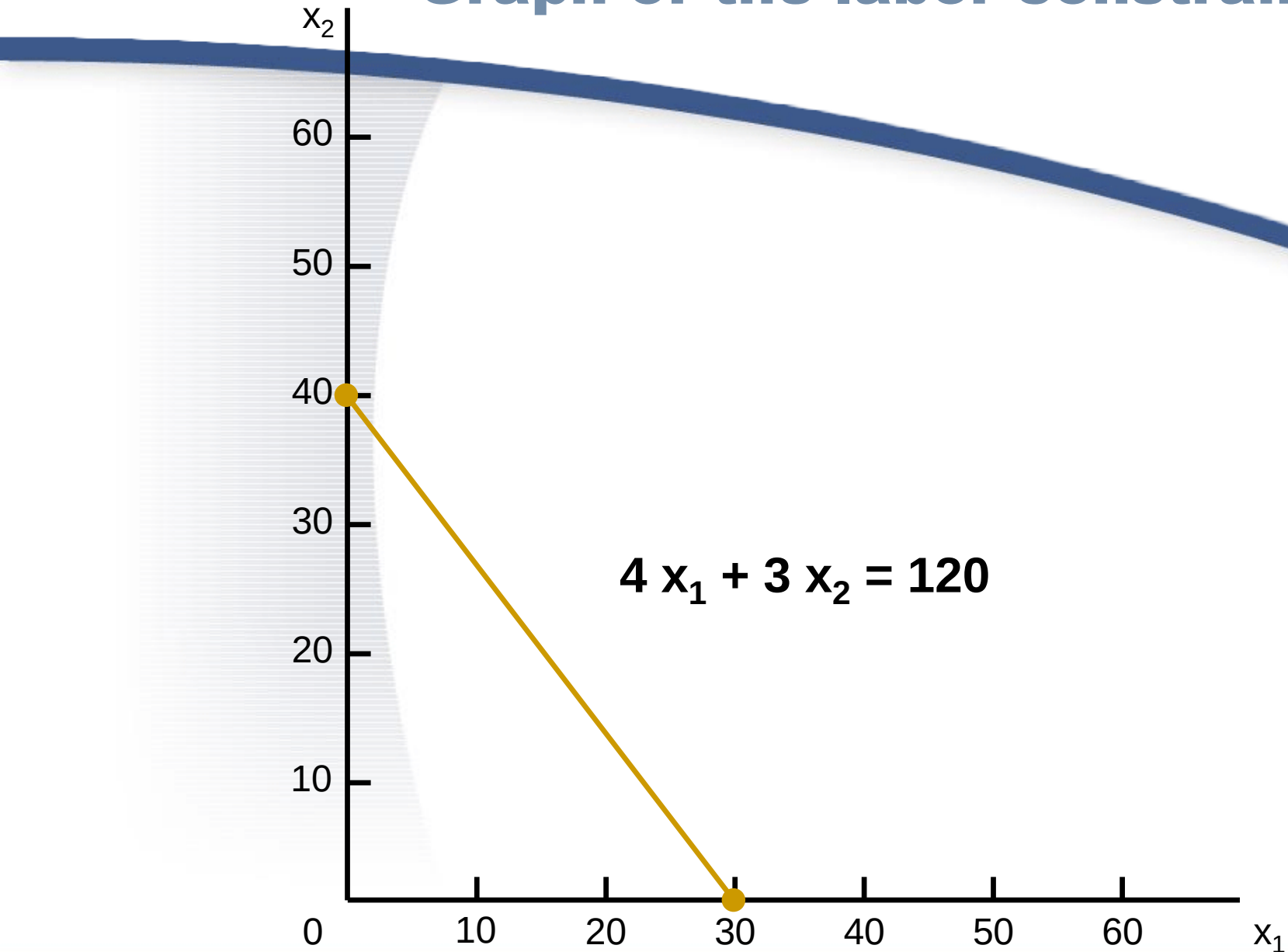
Graph of the labor constraint line



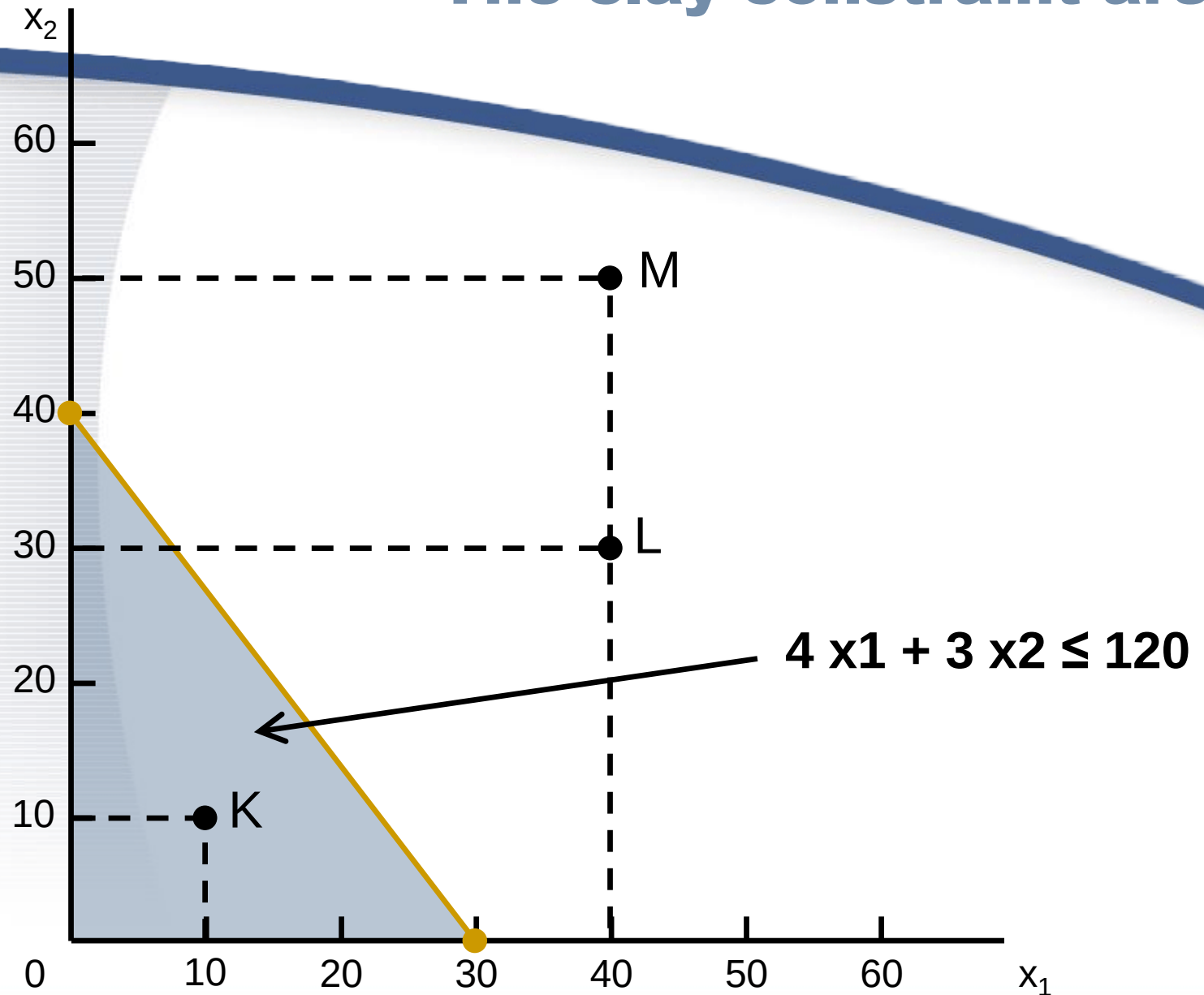
The labor constraint area



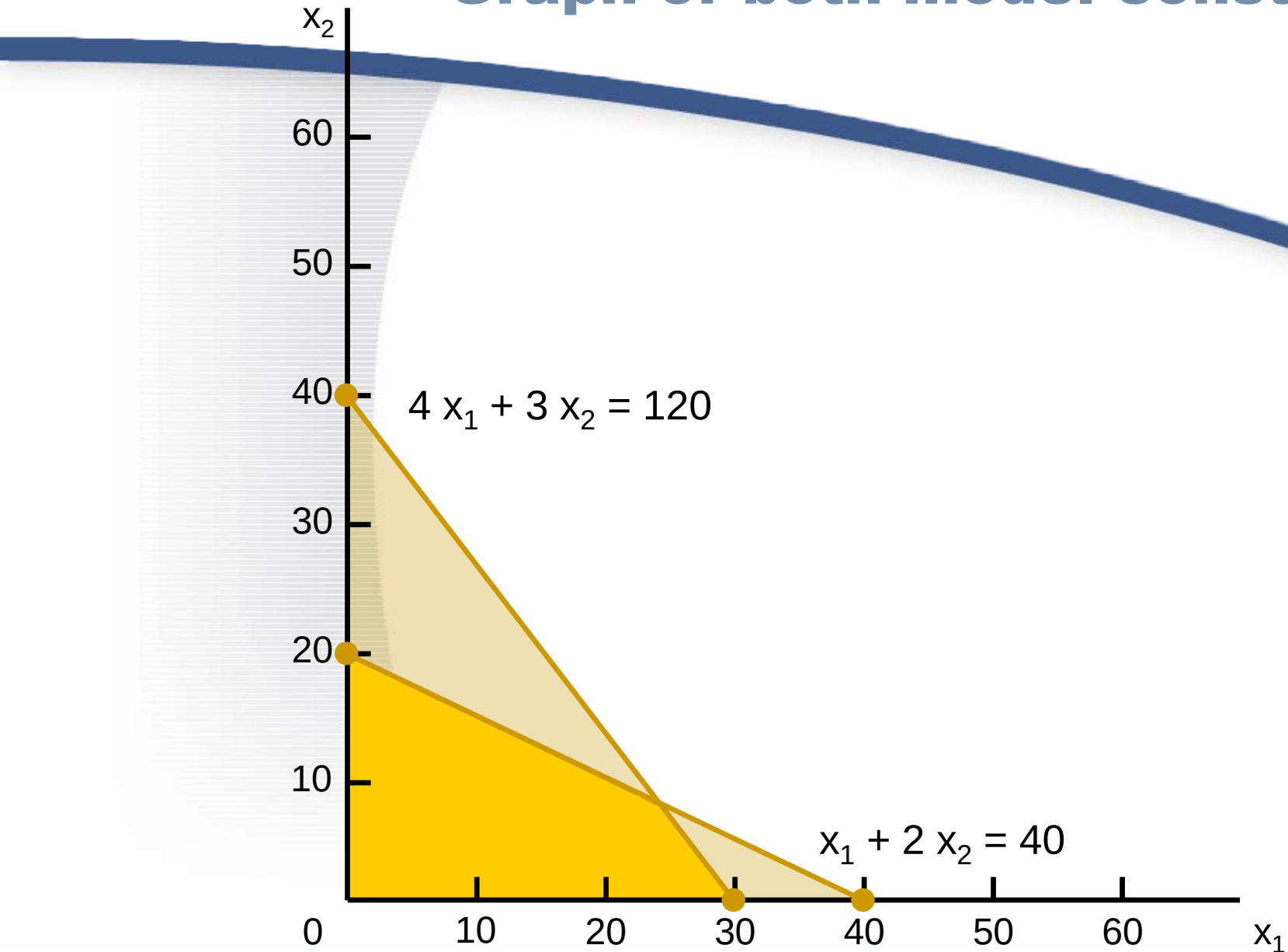
Graph of the labor constraint line



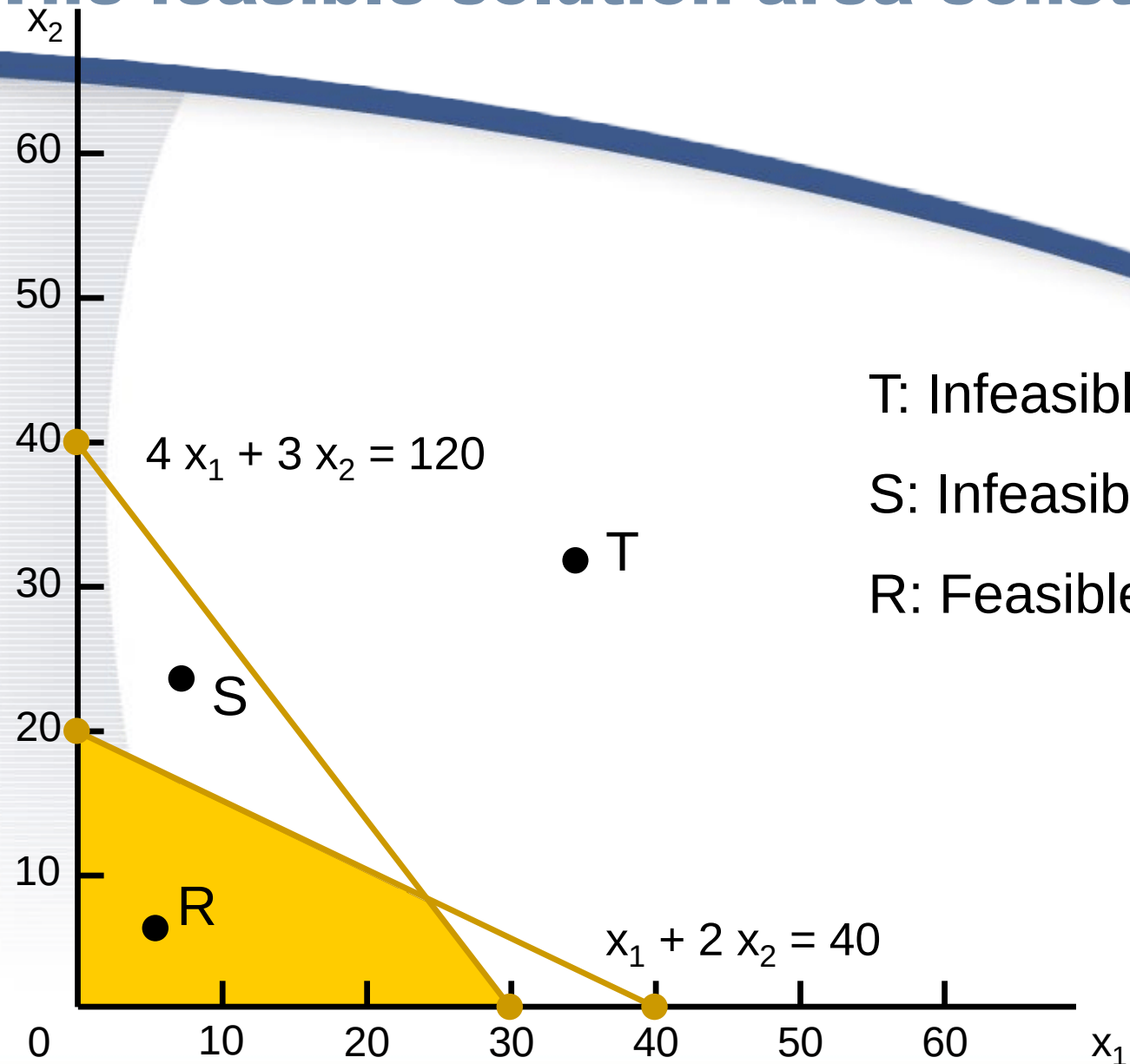
The clay constraint area



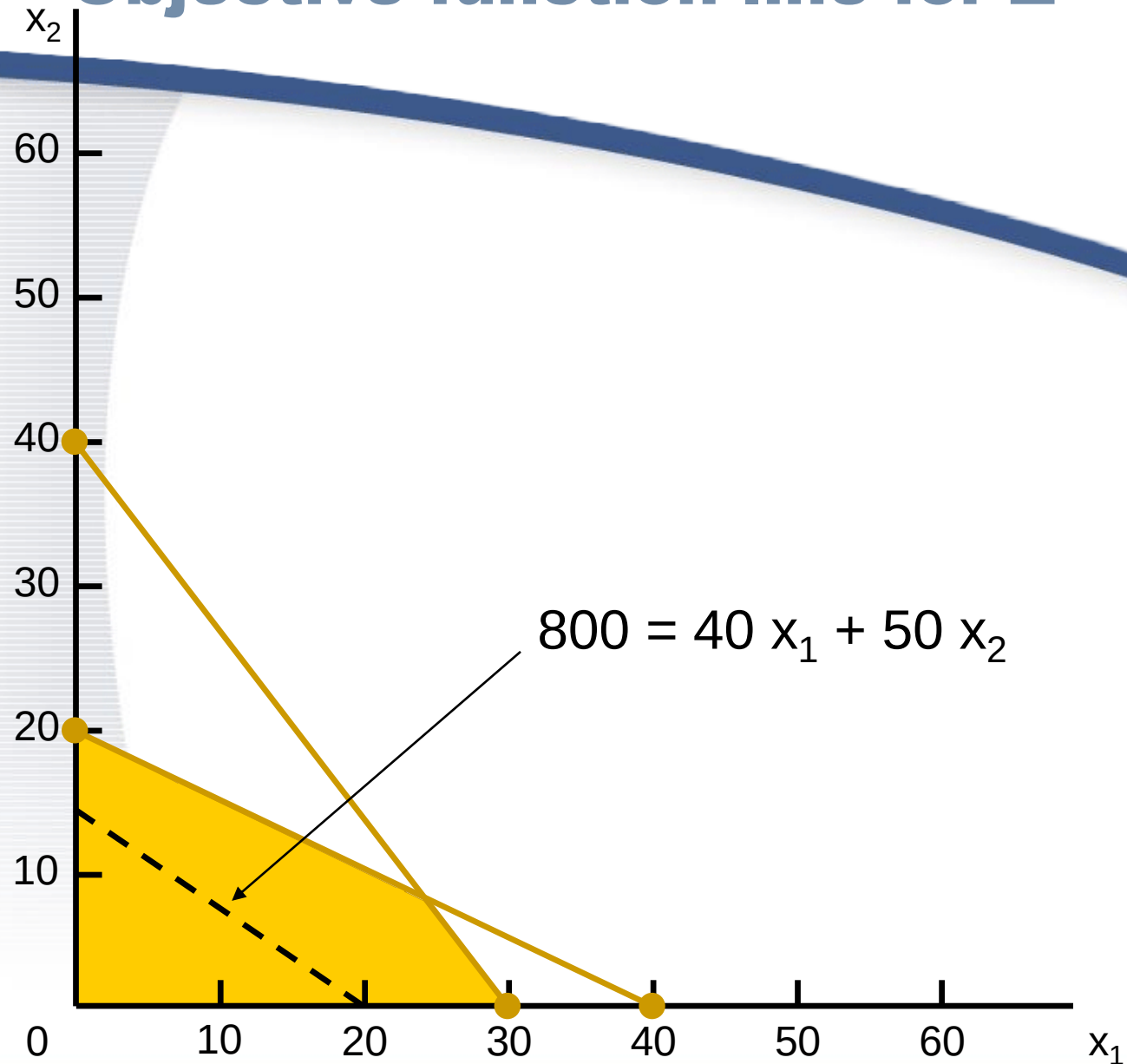
Graph of both model constraints



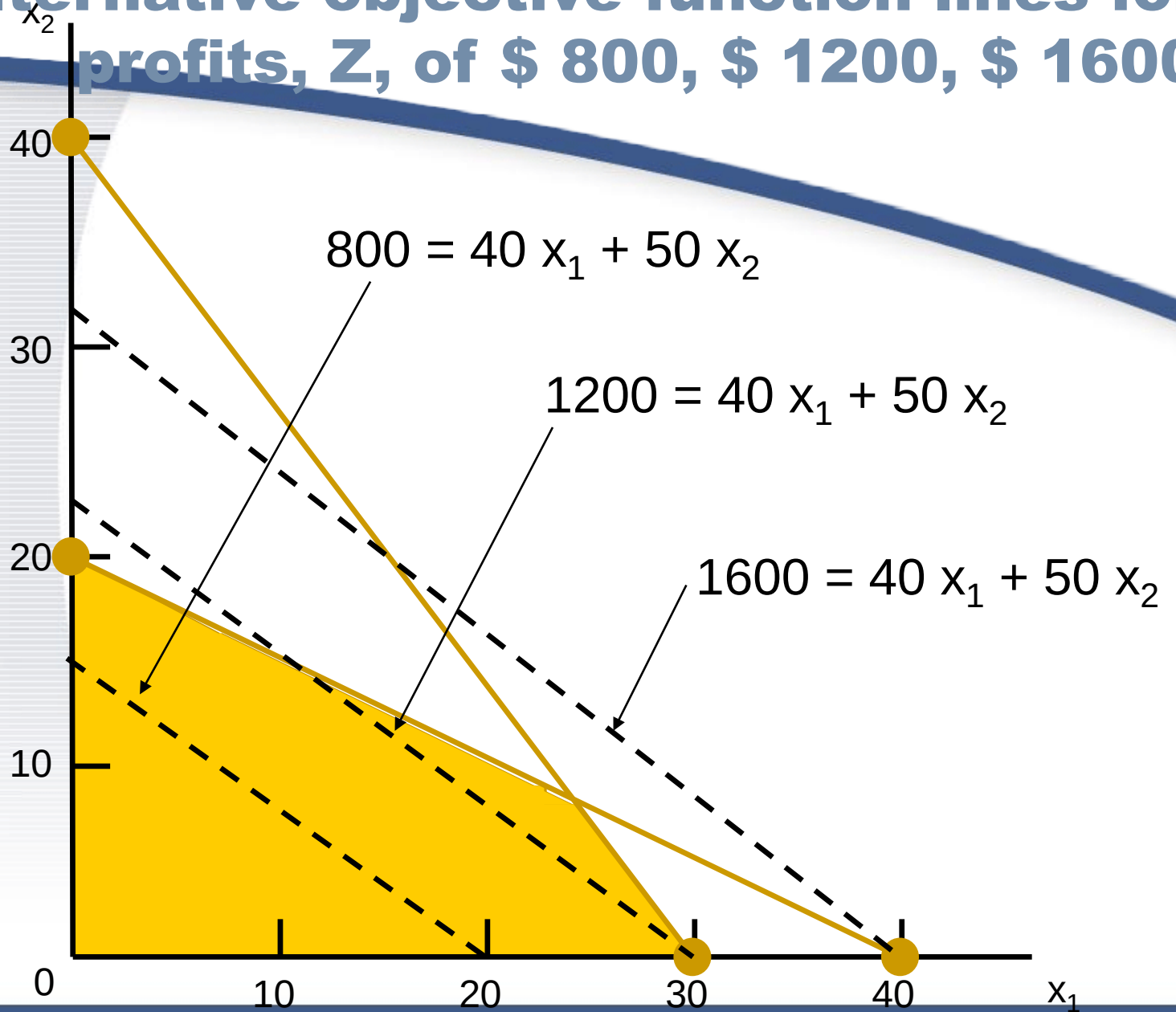
The feasible solution area constraints



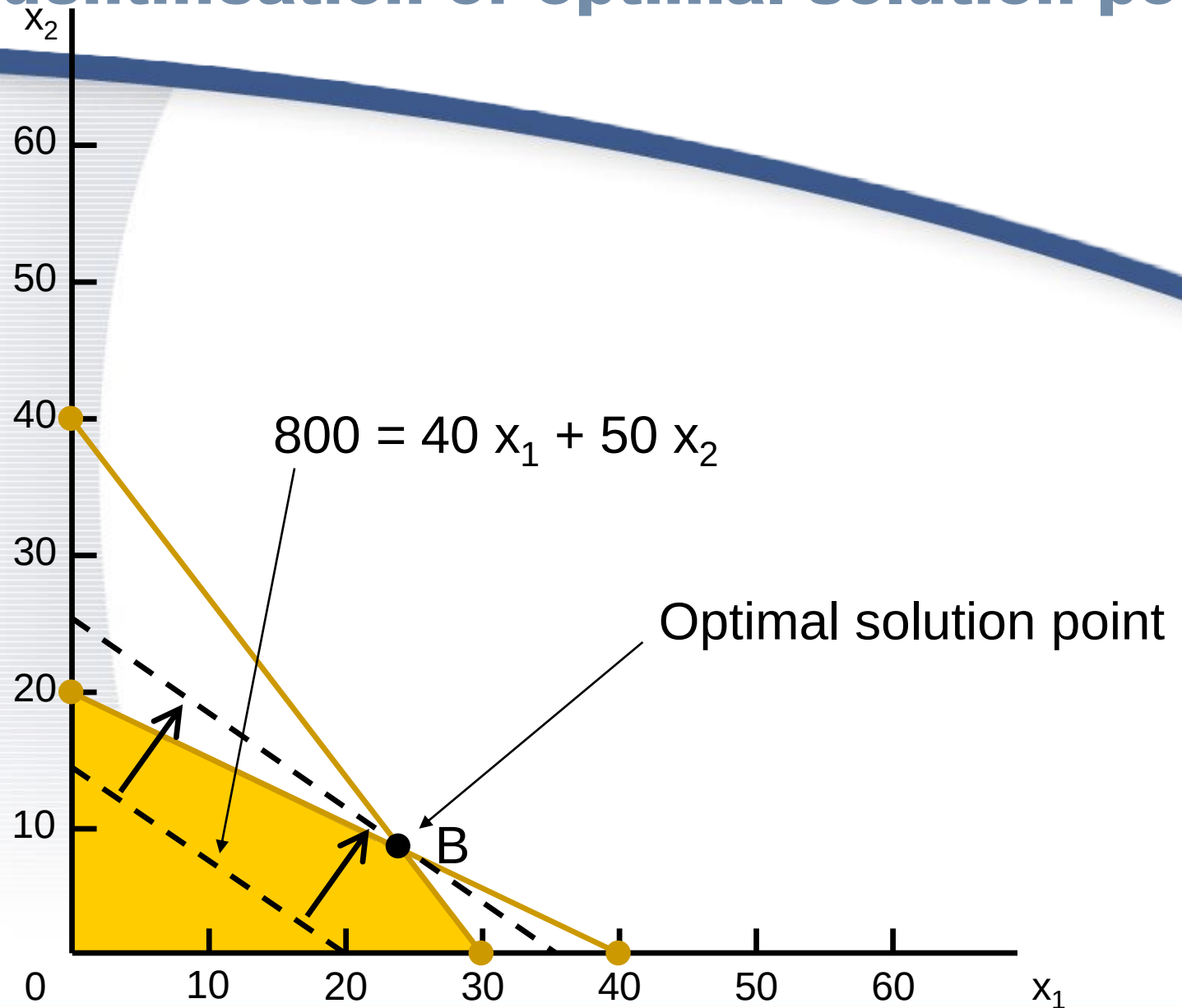
Objective function line for $Z = \$ 800$



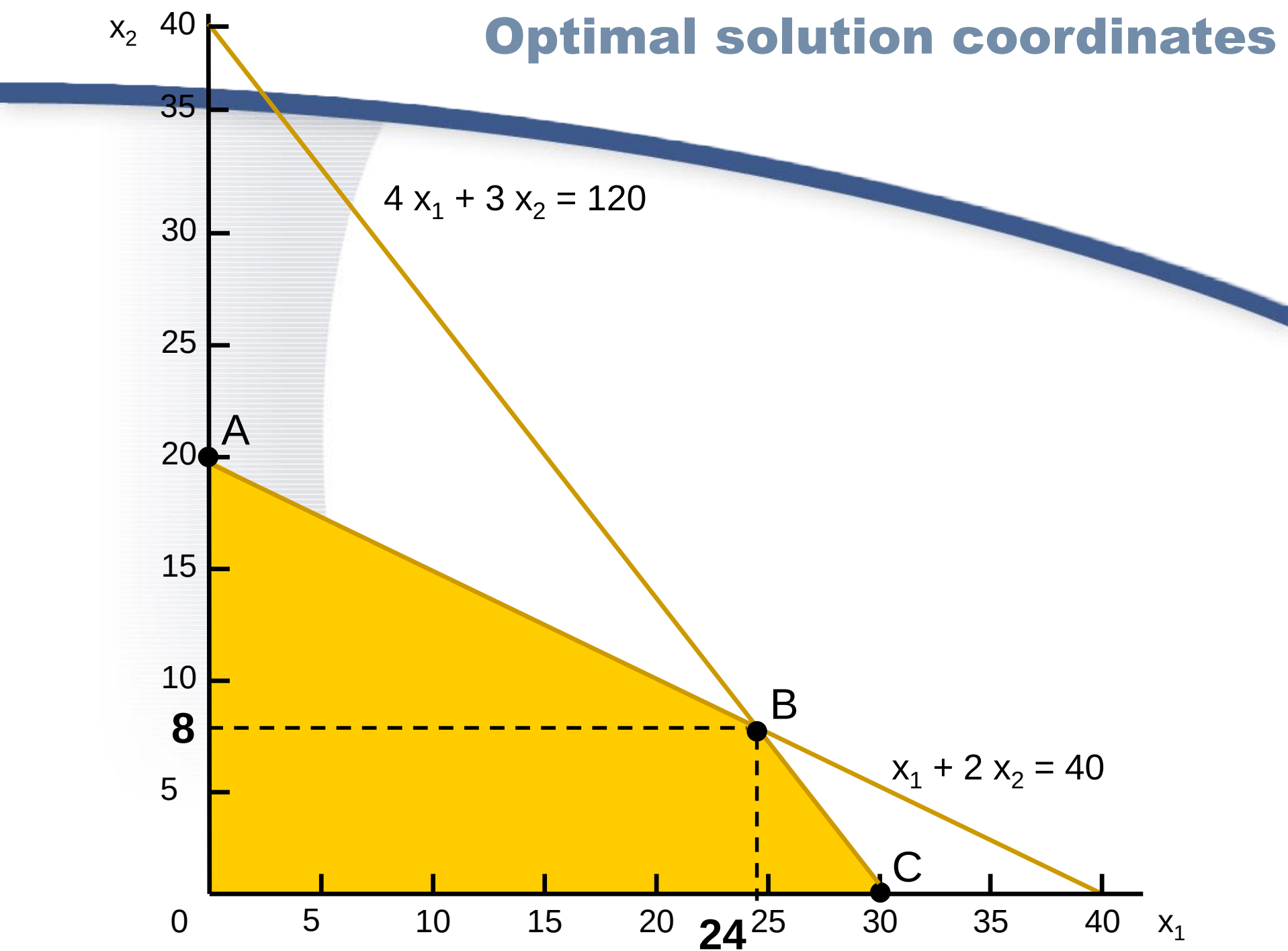
Alternative objective function lines for profits, Z , of \$ 800, \$ 1200, \$ 1600



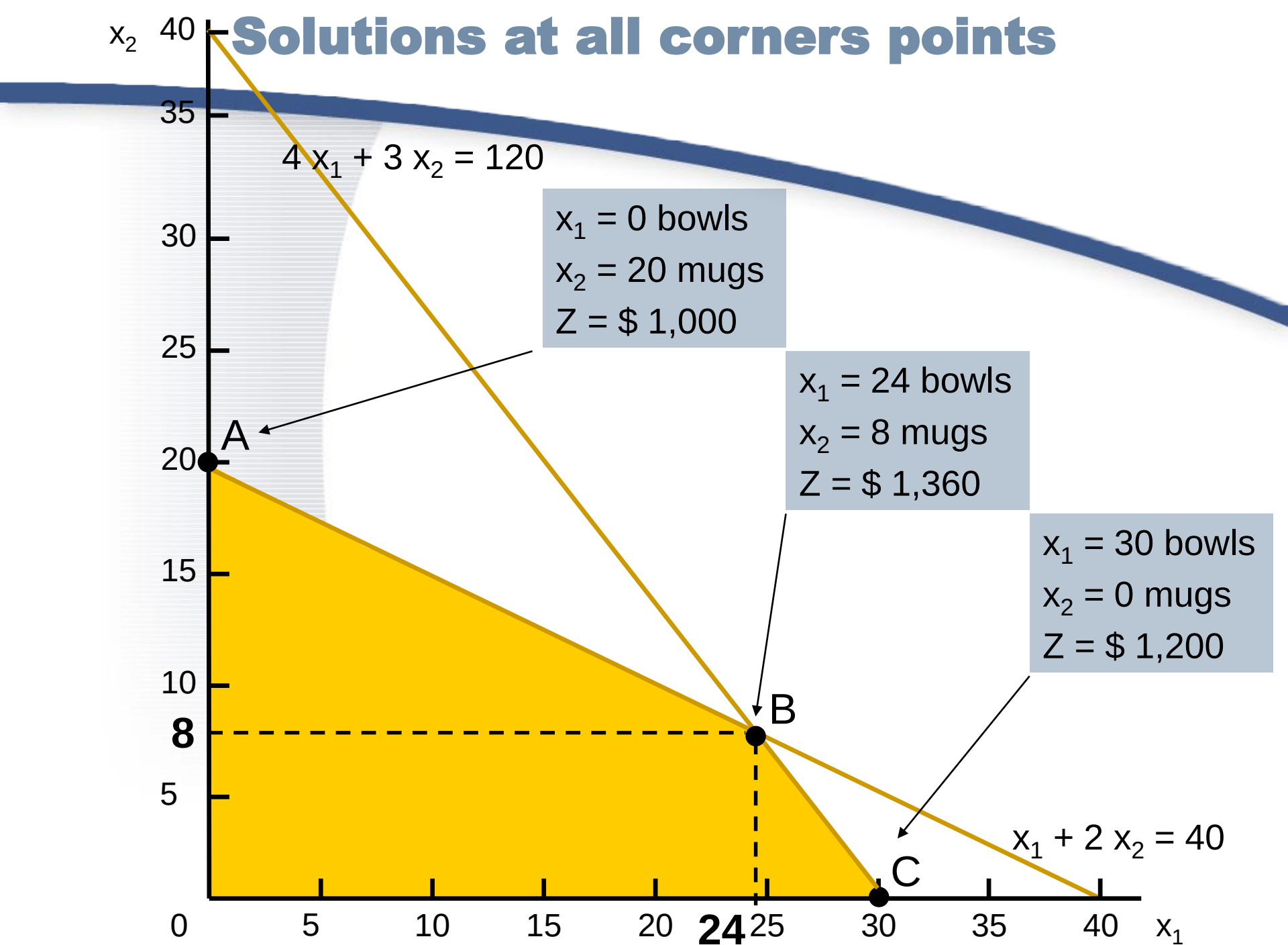
Identification of optimal solution point



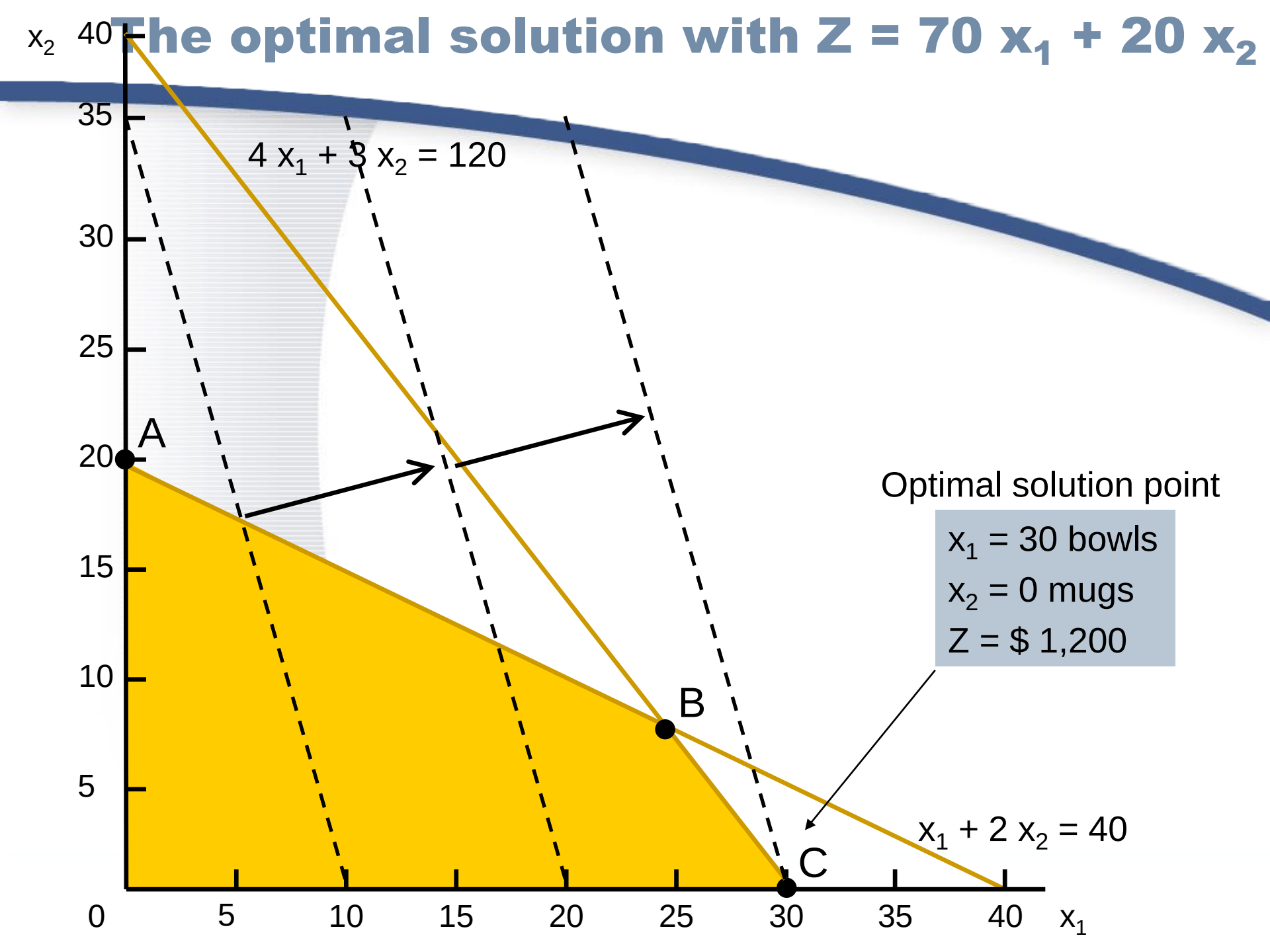
Optimal solution coordinates



Solutions at all corners points



The optimal solution with $Z = 70x_1 + 20x_2$



The steps for solving a graphical linear programming model are summarized next.

1. Plot the model constraints as equations on the graph; then, considering the inequalities of the constraints, indicate the feasible solution area.
2. Plot the objective function; then move this line out from the origin to locate the optimal solution point.
3. Solve simultaneous equations at the solution point to find the optimal solution values.

Or:

2. Solve simultaneous equations at each corner point to find the solution values at each point.
3. Substitute these values into the objective function to find the set of values that results in the maximum Z value.

A Minimization Model Example

The Farmer's Field

The farmer's field requires at least **16 kg. of nitrogen** and **24 kg. of phosphate**.

Super-gro costs \$6 per bag, and Crop-quick costs \$3.

The farmer wants to know how many bags of each brand to purchase in order to minimize the total cost of fertilizing.

Example: The Farmer's Field

Brand	Chemical Contribution	
	<i>Nitrogen</i> (kg./bag)	<i>Phosphate</i> (kg./bag)
Super-gro	2	4
Crop-quick	4	3

Linear Programming Model

$$\text{Minimize } Z = 6 x_1 + 3 x_2$$

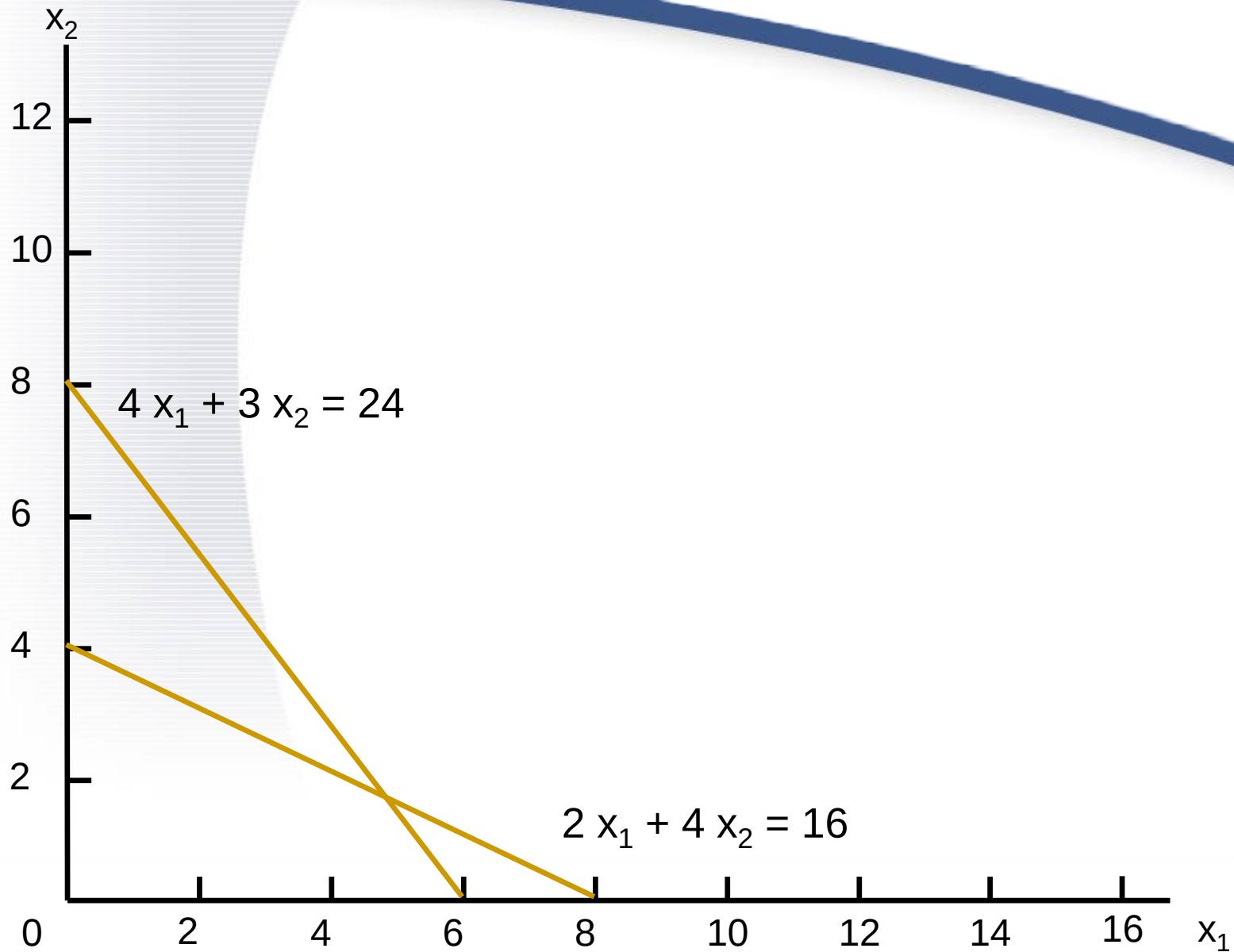
Subject to:

$$2 x_1 + 4 x_2 \geq 16$$

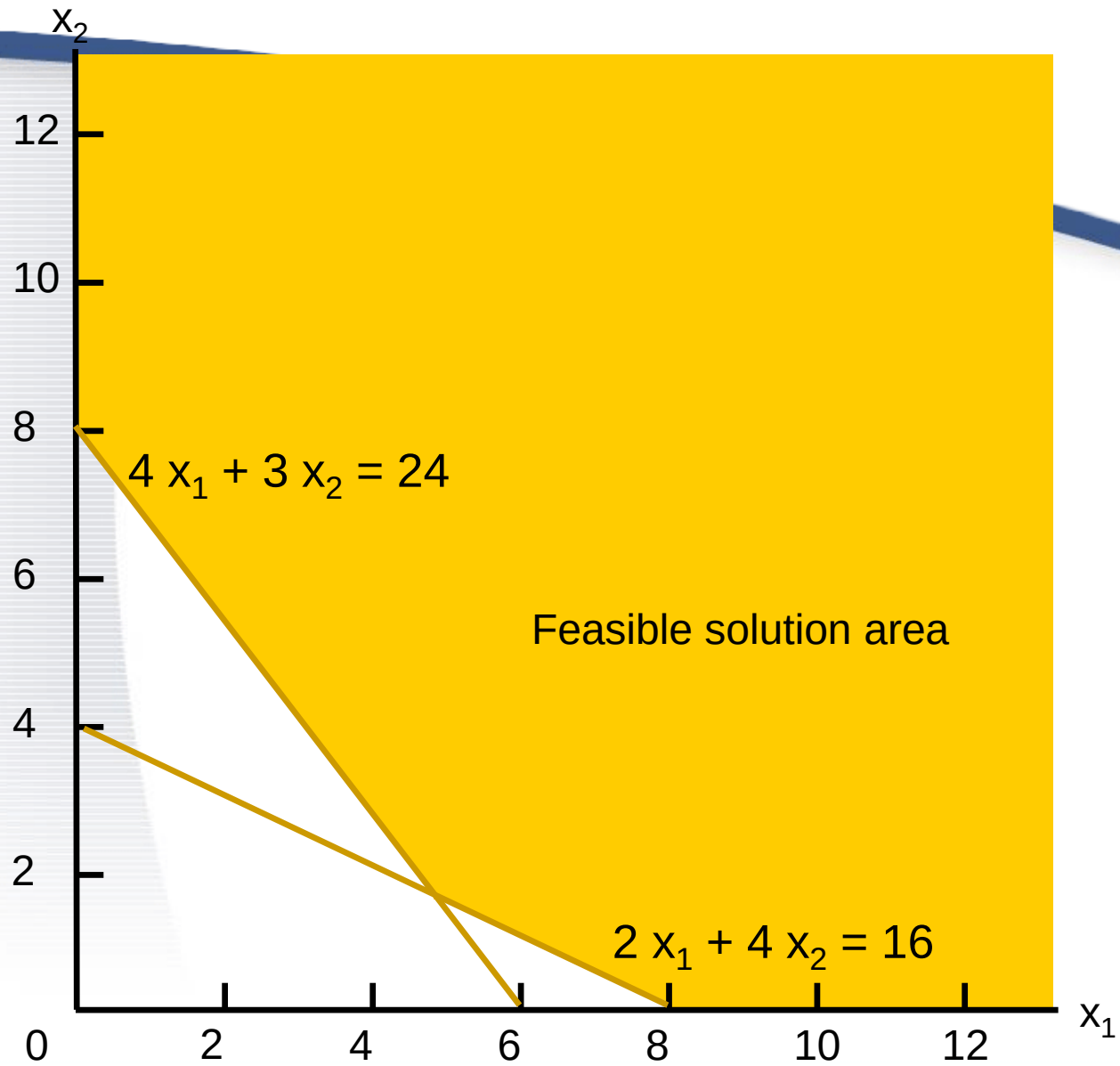
$$4 x_1 + 3 x_2 \geq 24$$

$$x_1, x_2 \geq 0$$

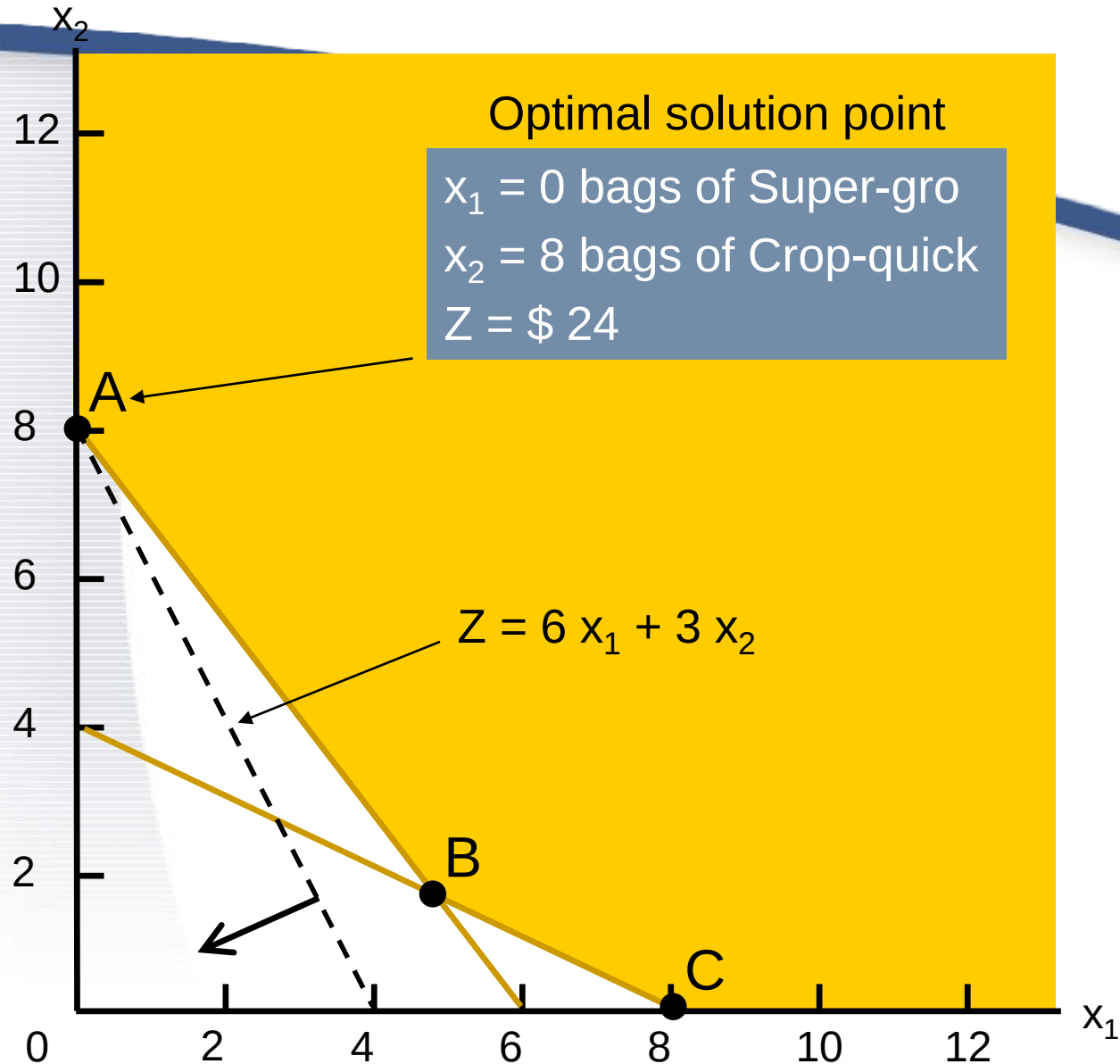
Constraint lines for fertilizer model



Feasible solution area



The optimal solution point

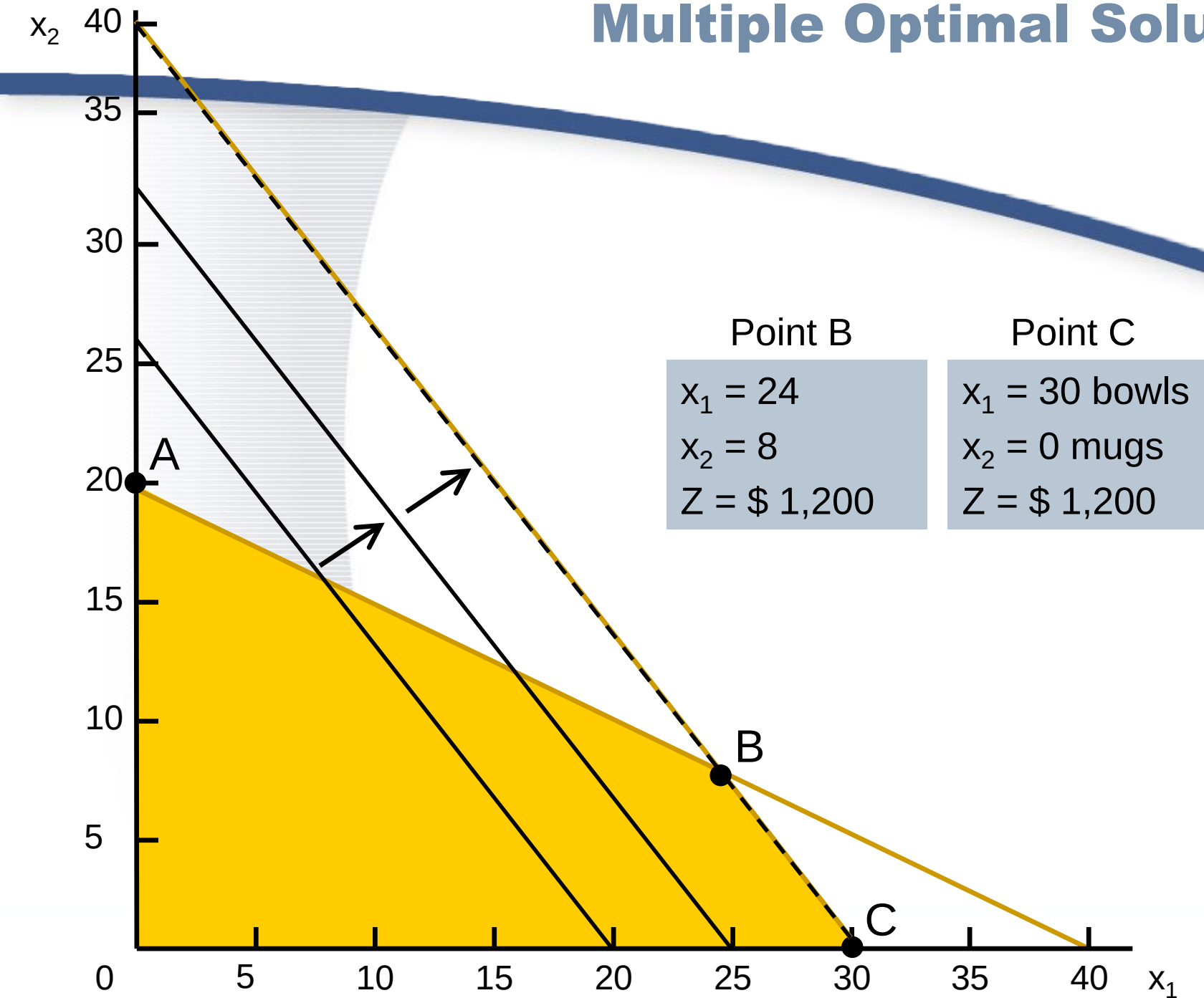




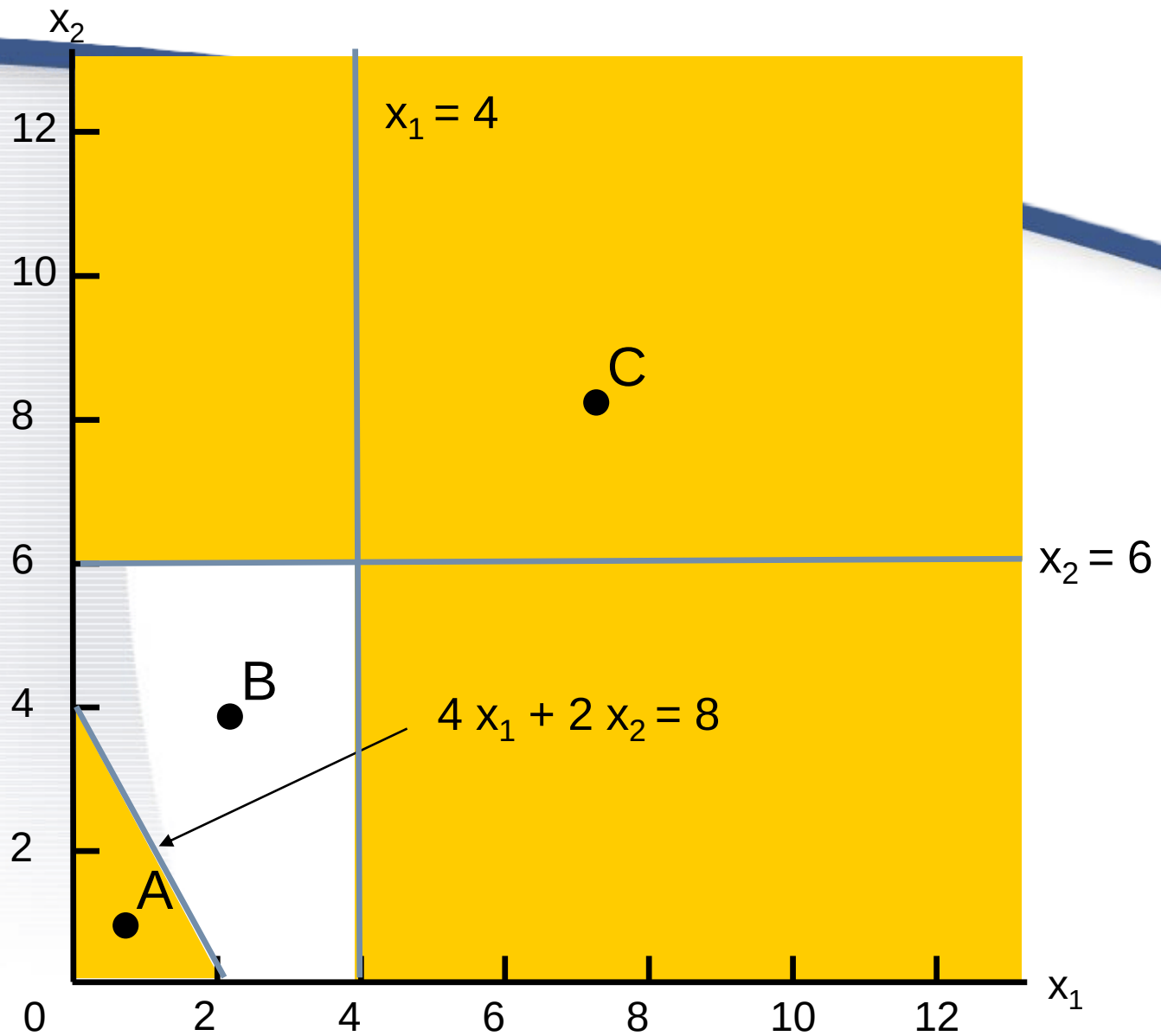
Irregular Types of Linear Programming Problems



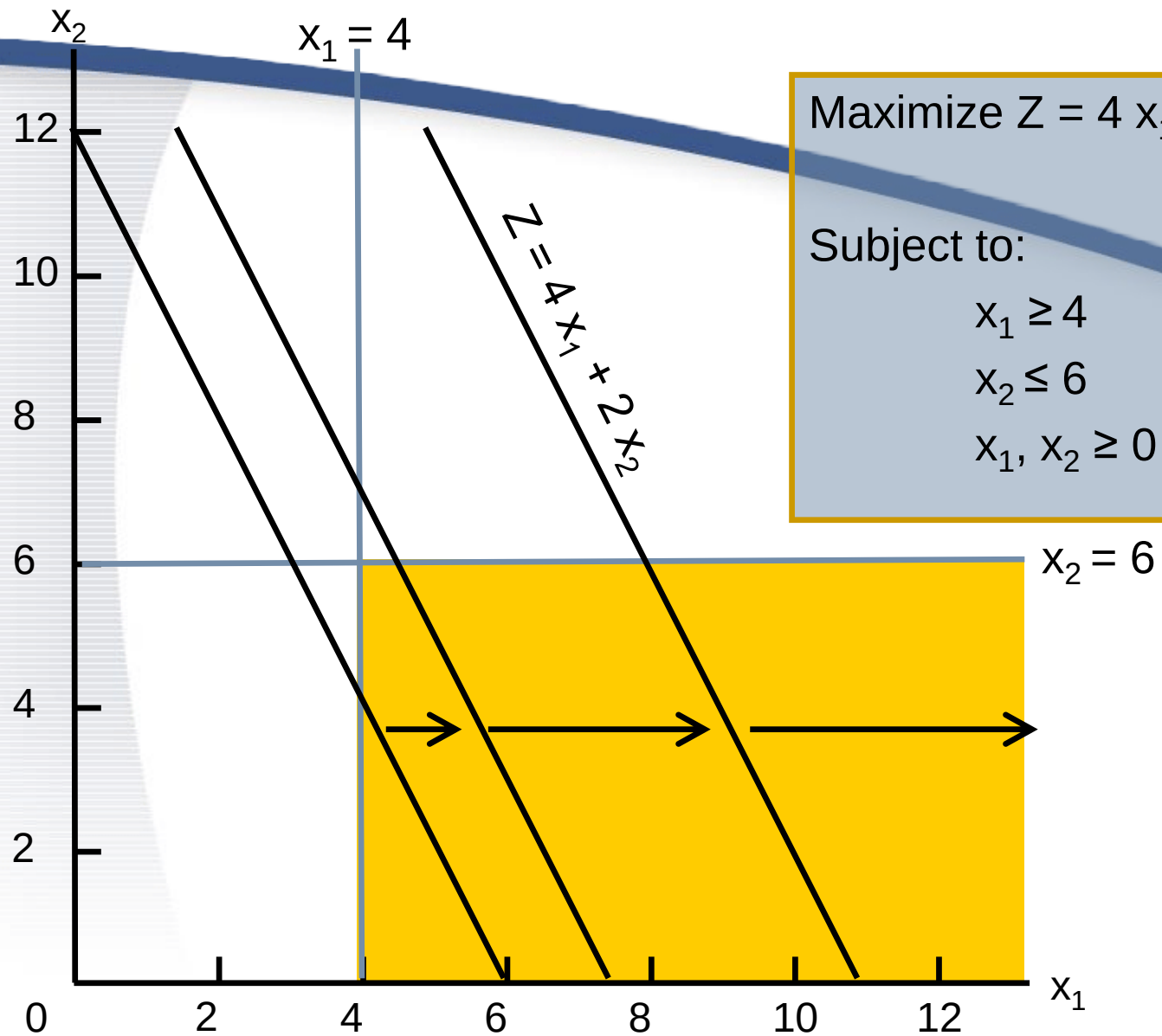
Multiple Optimal Solutions



An Infeasible Problem



An Unbounded Problem





Linear programming models exhibit certain common characteristics:

- An objective function to be maximized or minimized
- A set of constraints
- Decision variables for measuring the level of activity
- Linearity among all constraint relationships and the objective function